Casey Barton

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PROFILE

Applied-AI systems engineer based in Honolulu. I build the machinery that makes autonomous workflows trustworthy: evaluation harnesses that measure both model outcomes and agent/workflow quality — correctness, reproducibility, regressions, cost, latency — and guardrails with real veto power over agentic loops. My practice is modeling leverage through rigor: agentic workflows for design, train, evaluate, and improve cycles, with experiment readouts decided by evidence.

Three years running an evidence-grade agent control plane: typed state transitions (nominal / recoverable / hard-no-go), LLM-as-judge eval harnesses with explicit release thresholds, audit receipts on every run. I set technical direction for a 1,395-repository applied-AI estate — platform patterns (claim tracking, evidence boundaries, review gates) adopted beyond my own projects. Three years of intensive applied-AI systems work; the density of deployed eval and reliability systems is the counter-position to the 9+ year bar.

CORE COMPETENCIES

Engineering: Python · TypeScript · Node.js · SQL · Bash · REST APIs · Supabase · Postgres · Vercel · GitHub Actions (CI/CD)

Agentic ML Systems: evaluation harnesses (model outcomes + agent/workflow quality) · guardrails for autonomous/semi-autonomous workflows · error analysis & experiment readouts · LLM-as-judge · regression & latency benchmarking · structured experiment loops

Agents & Reliability: multi-agent orchestration · MCP protocol · LLM retrieval & structured outputs · fail-closed recovery · root-cause quarantine · observability with structured run states · audit receipts

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Evaluation-first delivery (job-app-helix): **989-test** agent control plane; evals with explicit release thresholds, gold sets and LLM-as-judge — the harness measures outcome quality and workflow quality (correctness, reproducibility, regressions, cost, latency), and the hard-failure path exits non-zero with CI asserting it.
- Evidence state machine as the trust mechanism: nominal / recoverable / hard-no-go — review gates and human-in-the-loop controls with real veto power; the engine raises instead of rendering when a bold claim isn't on the evidence allowlist. Guardrails that decide, not decorate.
- Failure analysis with root-cause quarantine instead of silent retries (apex-control-plane): SHA-256 transport envelopes, idempotency keys, circuit breakers, dead-letter capture, fail-closed startup — the reliability patterns that make semi-autonomous workflows trustworthy.
- Multi-agent orchestration (anthropic-agent-coordinator): deterministic DAG scheduler, **62/62** tests green; tool calls proposed for review, never executed blind — agentic workflows with typed state transitions and durable state that survives a failed run.

- Stateful retrieval (unified-memory-mcp, mcp-iceberg-mansion): Mem0, Supermemory, Supabase pgvector, Pinecone, and Neo4j federated behind one MCP interface — parallel fan-out, content-hash dedup; retrieval-grounded generation with provenance.
- Operate a **1,395-repository** applied-AI estate — **65 MCP connectors** integrated with claim tracking and idempotent automation; set technical direction adopted across projects (Forward-Deployed-Agentic-Powerhouse doctrine); **20+ production repositories** maintained under test and CI.

Chief Inspector 2024

Diamond Head Home Inspections — Honolulu, HI

- Led residential inspection operations: field assessments, defect documentation, and client reporting — measurement-first judgment from regulated, evidence-bound work.

EDUCATION

University of Hawaii at Manoa 2016

Bachelor of Science

AWS Cloud Institute 2025

Cloud Computing program (brief attendance)

Colorado State University

Undergraduate coursework

SELECTED PROOF

Job-App Helix — attributable campaign engine

- Composes assessments into campaign-level **GO / NO-GO** with immutable evidence models, stable finding codes, and fail-closed decisions — the eval harness that decides whether an agent change ships.
- Verification gates on every agent action: no agent completion claim without independent provider readback; recovery restricted to contingency evidence declared before reassessment.

GlacierEQ estate — agent systems portfolio

- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest; error analysis from evidence, not argument.
- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate — the compounding platform output of three years of shipped systems.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

Applied AI: LLM evaluation & eval harnesses · LLM-as-judge · multi-agent orchestration · RAG · prompt & context engineering · MCP protocol

Data & Infra: Supabase · Postgres · Vercel · GitHub Actions · REST APIs · experiment tracking discipline

Practice: evals with release thresholds · guardrails with veto power · root-cause quarantine · observability · mentoring through doctrine & playbooks

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment at Reddit, deep ML-modeling (training/fine-tuning) production depth, or certification. ~3 years intensive applied-AI systems work; 9+ year bar NOT rounded up.